

AE 18: Logistic regression introduction

! Important

- Open [RStudio](#) and create a subfolder in your AE folder called “AE-18”.
- Go to the [Canvas](#) and locate your AE-18 assignment to get started.
- Upload the `ae-18.qmd` and `framingham.csv` files into the folder you just created. The `.qmd` and PDF responses are due in Canvas. You can check the due date on the Canvas assignment.

Packages

```
library(tidyverse)
library(broom)
library(ggformula)
library(mosaic)
library(knitr)

heart_disease <- read_csv("framingham.csv") |>
  select(totChol, TenYearCHD) |>
  drop_na() |>
  mutate(high_risk = TenYearCHD) |>
  select(totChol, high_risk)
```

Data: Framingham study

This data set is from an ongoing cardiovascular study on residents of the town of Framingham, Massachusetts. We want to use the total cholesterol to predict if a randomly selected adult is high risk for heart disease in the next 10 years.

- `high_risk`:
 - 1: High risk of having heart disease in next 10 years
 - 0: Not high risk of having heart disease in next 10 years
- `totChol`: total cholesterol (mg/dL)

Exercise 1

Generate a plot, table of `high_risk`, and a plot to visualize the relationship between `high_risk` and `totChol`. Hint: none of these should be scatterplots.

Exercise 2

Generate a scatterplot of `totChol` vs. `high_risk`. Use `gf_lm()` to add a line to your plot. Do you think this is a good model? Why or why not?

Exercise 3

State whether a linear regression model or logistic regression model is more appropriate for each scenario.

1. Use age and education to predict if a randomly selected person will vote in the next election.
2. Use budget and run time (in minutes) to predict a movie's total revenue.
3. Use age and sex to calculate the probability a randomly selected adult will visit St. Lukes in the next year.

Exercise 4

Using the table you generated in Exercise 1:

0. Based on our data, what is considered “success”?
1. What is the probability a randomly selected person in the study is **not** high risk for heart disease in the next 10 years?
2. What are the **odds** a randomly selected person in the study is **not** high risk for heart disease in the next 10 years?
3. What are the **log-odds** a randomly selected person in the study is **not** high risk for heart disease in the next 10 years?

Submission

! Important

To submit the AE:

- Render the document to produce the PDF with all of your work from today's class.
- Upload your QMD and PDF files to the Canvas assignment.